BevOS: Software Requirements Specification

Lab 2 – Draft 1

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1 Introduction

This document presents the software requirements and specifications for BevOS, a centralized workflow management platform for the beverage manufacturing industry. It outlines the system's scope, major functions, user characteristics, and technical considerations. The following sections describe the requirements that will guide the development of the BevOS prototype.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to establish a common understanding of what the BevOS prototype must accomplish. It provides the development team and stakeholders with a reference for evaluating system functionality, identifying development priorities, and verifying that the completed prototype meets its defined requirements.

1.2 Scope

BevOS is a centralized workflow management platform designed to improve coordination and communication within the beverage manufacturing industry. The system aims to replace fragmented processes involving spreadsheets, emails, phone calls, and disconnected software with a single platform that connects brands, suppliers, distributors, co-packers, logistics providers, and internal teams.

BevOS will support activities such as product requests, quote management, approvals, purchasing, production tracking, logistics coordination, document storage, and reporting. By centralizing these processes and automating repetitive tasks, the system aims to reduce communication delays, approval bottlenecks, and operational inefficiencies. The BevOS prototype will demonstrate how these capabilities can work together to improve workflow visibility and coordination throughout the beverage manufacturing process.

1.3 Definitions, Acronyms, and Abbreviations

- **Approval Workflow** - A process used to review and approve operational tasks before they move to the next stage.
- **BevOS** - A centralized workflow-management platform designed for beverage manufacturing operations.
- **Client** - An organization or individual purchasing services or products from another participant in the manufacturing process.
- **Co-Packer** - A third-party company that manufactures and packages beverages for another company.
- **Core Services** - The shared BevOS system software layer that processes requests, applies system rules, and coordinates functional modules, and manages data exchange.
- **Dashboard** - A visual interface displaying operational information and performance metrics.
- **Distributor** - An organization responsible for delivering products from manufacturers to retailers or customers.
- **ERP (Enterprise Resource Planning)** - Software that integrates core business processes into a unified system.
- **Logistics** - The planning and coordination of transportation and delivery activities.
- **MFCD (Major Functional Components Diagram)** - A diagram illustrating the primary hardware and software components of the system.
- **Nice Package Company** - The beverage-industry client and domain partner for whom BevOS is being developed.
- **PostgreSQL** - An open-source relational database management system used to store application data.
- **Role-Based Access Control** - A security method that grants system permissions according to the user's assigned role and responsibilities.
- **Single Source of Truth** - A trusted, centralized set of current information used consistently by authorized stakeholders.

- **SRS (Software Requirements Specification)** – A document defining the functional and nonfunctional requirements of a software system.
- **Stakeholder** - A person or organization that participates in or is affected by a beverage-manufacturing workflow.
- **Workflow** - A sequence of business activities completed to achieve a specific objective.

1.4 References

- Grand View Research. (2024). Beverage Packaging Market Size, Share & Trends Analysis Report.
- Kennedy, T. (2026). CS410 Professional Workforce Development.
- Wilson, T. (2026). BevOS Presentation / Nice Package Co. Guest Presentation.
- Zimmerman, S. (2026) How the food and beverage industry is rethinking manufacturing investments. Food Dive.

1.5 Overview

The remainder of this document is organized into two main sections. Section 2 provides an overall description of BevOS, including its product perspective, main product functions, user characteristics, constraints, and assumptions and dependencies. Section 3 presents specific software requirements, describing the system's functional capabilities, interfaces, performance expectations, and other requirements necessary for the development and evaluation of the BevOS prototype.

2 Overall Description

This section provides an overview of BevOS and its role in coordinating beverage manufacturing operations. It describes the system's relationship to existing processes, primary functions, intended users, and technical considerations that may influence its development. The following subsections establish the context for the specific software requirements presented in Section 3.

2.1 Product Perspective

BevOS is a web-based application designed to centralize workflow management within the beverage manufacturing industry. The system will provide a single platform for brands, distributors, suppliers, and internal teams to coordinate activities that are currently managed through separate spreadsheets, emails, phone calls, and disconnected software systems.

As illustrated in Figure 1, BevOS is organized into several interconnected software components. Users access the system through the BevOS User Interface, which communicates with BevOS Core Services to process requests, enforce system rules, and coordinate the application's functional modules. These modules include Quotes, Communications, Production, Logistics, Document Storage, and Reports, A PostgreSQL relational database provides persistent storage for the information managed throughout the system.

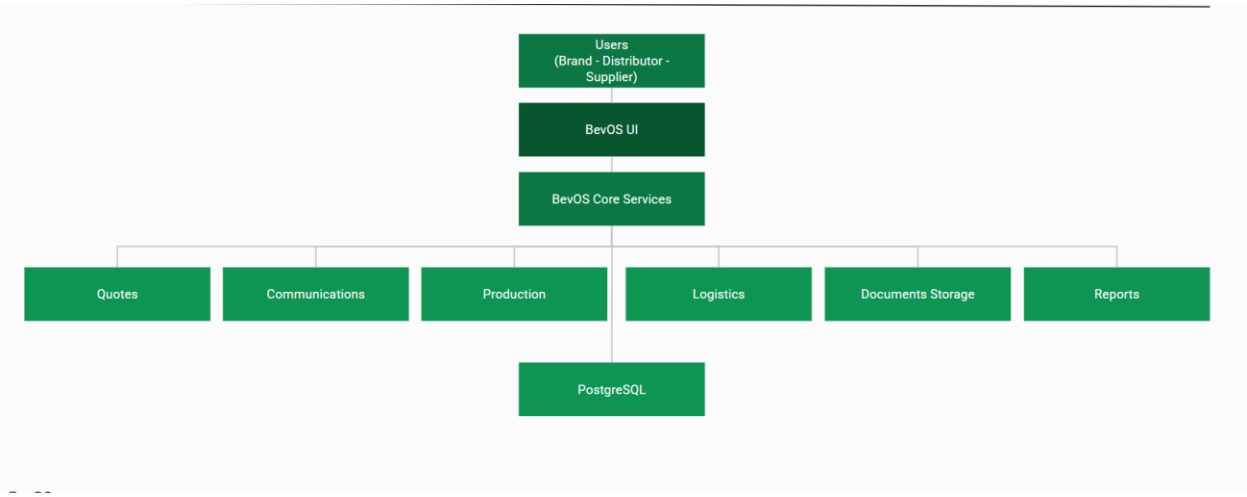


Figure 1: Major Functional Components Diagram

BevOS will operate through standard web browsers and will not require specialized hardware for end users. The application will rely on server infrastructure to host its interface, services, and database, although the specific production hosting environment has not yet been determined.

2.2 Product Functions

BevOS provides a centralized platform for managing and coordinating activities throughout the beverage manufacturing process. The system integrates six primary functional modules: Quotes, Communications, Production, Logistics, Document Storage, and Reports. These modules work together to improve operational visibility, streamline workflows, and reduce the need for disconnected tools and manual communication.

The Quotes module supports quote calculations and negotiation activities, while the Communications module facilitates information sharing among users. The Production module supports production tracking and approval workflows, and the Logistics module coordinates shipping and delivery activities. Document Storage provides a centralized location for managing operational documents, while the Reports module allows users to access information about workflow activities and operational performance.

The BevOS prototype will demonstrate a subset of the functionality intended for the real-world product. Approval tracking, production tracking, and logistics visibility are included in the protype's planned capabilities, with simulated data and supporting components used where necessary. Other functions, such as document management and non-target workflow modules, will have more limited implementations. Table 1 compares the capabilities of the proposed real-world product with those planned for the prototype.

Feature	**RWP BevOS**	**Prototype**
Centralized Workflow	Complete simulation of business	Stubs for non-target modules
Beverage Industry Focus	Initially within Adaptable Framework	Assumed sole focus
Approval Tracking	Complete	Complete
Production Tracking	Complete	Complete, drivers where needed
Logistics Visibility	Complete	Complete, synthetic data
Document Management	Technical and Client/Legal Consideration	Technical implementation only

Table 1: Real World vs. Prototype

2.3 User Characteristics

BevOS is designed for users involved in various stages of the beverage manufacturing process, including brands, suppliers, distributors, co-packers, logistics providers, and internal operations teams. Each user group has different responsibilities and requires access to specific information and system functions. Role-based access control will allow users to interact with the features and data relevant to their responsibilities.

Brands and clients will use BevOS to submit product requests, review quotes, approve artwork, and monitor the progress of their orders. Suppliers and co-packers will use the platform to coordinate materials, respond to requests, and manage production-related activities. Distributors and logistics providers will access information related to shipping, delivery schedules, and order

fulfillment. Internal operations teams will oversee workflows, manage approvals, track production activities, and coordinate communication among stakeholders. Users are expected to have basic computer skills and familiarity with web-based applications. BevOS is intended to support users with different levels of technical experience, so the interface should provide clear navigation and straightforward access to relevant workflow information.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A